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Subject: Cleansers Chemical ingredients | Next step

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ALCON,

Just want to keep everyone up to speed as new information comes. There is a way around the VOC compliance issue and that is BOV (Bag on Valve). In this type of system the pure cleaning solution is in the can in a pouch/bag and it is then pressurized with air(no hazardous propellants, eco-friendly), so VOC would no longer be a factor.

This Company Nyco Products (might be a good company if we start large scale production) put me on to BOV and to MB industries where Mark(retired army guy)can absolutely help us fill the cans and test them. So if I can provide him with the chemicals (premixed) he can fill the cans.

Below are some formulation recipes starting from the more basic (F1) and getting more complex by F4. Apollo, I don't know if this is something within Dave Zuiderveens realm but if you could look into that and I will look into additional ways to source these chemicals on a small scale for now.

Below are four different formulation recipes, I really want to start with peppermint as it is a huge maggot, bug, spider deterrent. I will have to source these chemicals and mix myself at home and then pack them and send them to MB industries for him to send back after he fills them, and THEN we can get to the real testing.

Formula 1 – Basic Peppermint Foaming Cleanser

Goal: simplest, eco-friendly foamer.

Ingredients Needed:

- Decyl Glucoside – 1 L (~\$30–60)

- Cocamidopropyl Betaine – 500 mL (~\$25–50)
- Peppermint Essential Oil – 2 oz (~\$10–20)
- Polysorbate-20 (solubilizer) – 250 mL (~\$10–20)
- Phenoxyethanol (or other preservative) – 50 g (~\$20–40)
- Distilled Water – 1–2 L (<\$10)

Approx Cost: ~\$120–180

Formula 2 – Mid-Range (More Cleaning Power)

Goal: better grease cutting and stability.

Ingredients Needed:

- Formula 1 items plus:
- Sodium Olefin Sulfonate (AOS) – 500 g (~\$20–40)
- Sodium Citrate (builder) – 250 g (~\$10–15)

Approx Cost: ~\$150–220

Formula 3 – Mid-Advanced (Balanced Surfactant System)

Goal: blend of mild + strong surfactants, chelation, foaming stability.

Ingredients Needed:

- Formula 2 items plus:
- GLDA (chelating agent) – 250 g (~\$20–40)

- Propylene Glycol (solvent, stabilizer) – 500 mL (~\$15–25)

Approx Cost: ~\$180–250



Formula 4 – Advanced Complex Cleanser

Goal: stable foaming, fragrance solubilized, antimicrobial preservative system.

Ingredients Needed:

- Formula 3 items plus:
- Caprylyl/Capryl Glucoside (secondary solubilizer) – 250 g (~\$15–25)
- Ethylhexylglycerin (co-preservative) – 50 g (~\$15–25)
- Sodium Bicarbonate (buffer/foam booster) – 500 g (~\$10–15)

Approx Cost: ~\$200–300



Core Equipment (one-time buy, works for all formulas)

- Graduated Cylinders / Glass Beakers – \$20–40
- Precision Scale (0.01 g accuracy) – \$25–50
- pH Meter – \$30–80
- Magnetic Stirrer w/ Hot Plate (optional but helpful) – \$50–120
- Safety Gear (gloves, goggles, apron) – \$25–50

One-time Cost: ~\$150–250

Very respectfully,

Jacob Bevier

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